

Molecular imaging in oncology: Current impact and future directions

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Abstract: The authors define molecular imaging, according to the Society of Nuclear Medicine and Molecular Imaging, as the visualization, characterization, and measurement of biological processes at the molecular and cellular levels in humans and other living systems. Although practiced for many years clinically in nuclear medicine, expansion to other imaging modalities began roughly 25 years ago and has accelerated since. That acceleration derives from the continual appearance of new and highly relevant animal models of human disease, increasingly sensitive imaging devices, high-throughput methods to discover and optimize affinity agents to key cellular targets, new ways to manipulate genetic material, and expanded use of cloud computing. Greater interest by scientists in allied fields, such as chemistry, biomedical engineering, and immunology, as well as increased attention by the pharmaceutical industry, have likewise contributed to the boom in activity in recent years. Whereas researchers and clinicians have applied molecular imaging to a variety of physiologic processes and disease states, here, the authors focus on oncology, arguably where it has made its greatest impact. The main purpose of imaging in oncology is early detection to enable interception if not prevention of full-blown disease, such as the appearance of metastases. Because biochemical changes occur before changes in anatomy, molecular imaging—particularly when combined with liquid biopsy for screening purposes—promises especially early localization of disease for optimum management. Here, the authors introduce the ways and indications in which molecular imaging can be undertaken, the tools used and under development, and near-term challenges and opportunities in oncology.

Keywords: magnetic resonance imaging (MRI), nuclear medicine, optical imaging, positron emission tomography (PET), single-photon emission computed tomography (SPECT), theranostics

Introduction

On its website, the Society of Nuclear Medicine and Molecular Imaging defines *molecular imaging* as a type of medical imaging that provides detailed pictures at the molecular and cellular levels of what is occurring inside the body.¹ This accurate, but staid, definition belies the immense progress researchers and clinicians have made over the past 2 decades in applying the principles of molecular imaging across several fields, from basic and translational science through state-of-the-art patient diagnosis and therapy. Fundamentally, molecular imaging allows for the visualization of biochemical processes and patterns of target localization that are invisible at the anatomic imaging level.

Although endogenous image contrast can be leveraged or induced within tissues,^{2,3} much of molecular imaging requires administration of an imaging agent, usually intravenously, which interacts with a targeted environment to uncover biological pathways. Because a hallmark of molecular imaging is lack of perturbation of the cell, environment, or process under study, the imaging agents often serve as tracers, with no effect on the entity they are designed to measure. Tracers can be molecules or analogs of molecules that participate in metabolic pathways or they can be targeted to serve as substrates for or bind to specific enzymes,

receptors, antigens, or transporters. In many scenarios, the tracer will be radiolabeled, ie, with a radionuclide, although, as discussed below, this is not always the case. A second component necessary for molecular imaging is appropriate hardware—a sensor or scanner that can detect the tracer and translate that detection into spatial information. Optimized molecular imaging approaches will have a high-affinity tracer for a pathway or target that is near-uniquely present in the process of interest, as well as a scanner with high sensitivity and high spatial, contrast, and temporal resolution.

In the current review, we focus on selected, common imaging modalities and examples that highlight molecular imaging in oncology. Specifically, we detail techniques in optical and near-infrared (NIR) imaging, magnetic resonance imaging (MRI), and nuclear medicine techniques, including single-photon emission computed tomography (SPECT) and positron emission tomography (PET) (Table 1).^{4,119} We also provide specific examples from translational science and cancer clinical care of the utilization of molecular imaging, with a particular focus on the use of these methods to guide and improve patient management. Finally, we delineate the challenges faced by the field and the potential benefits of overcoming them.

Modalities

A comprehensive description of all molecular imaging modalities is beyond the scope of this review, with several valuable reviews having recently appeared.^{120–122} Accordingly, we endeavored to highlight a subset of the most commonly used modalities and their relative advantages and disadvantages. Key aspects of these modalities are listed in Table 1.

Optical Techniques

Optical imaging is primarily a preclinical tool, although its extensive use in molecular imaging in small animal models of cancer merits discussion here. In many modern early phase clinical trials, aspects of the biological justifications for many of the agents being investigated have been preclinically evaluated with optical imaging techniques. Optical imaging subsumes multiple submodalities, including bioluminescence imaging (BLI), fluorescence, and chemiluminescence.⁴ BLI, first reported by Contag and colleagues, enabled the ability to follow cellular activity, including gene expression, in living animals.¹²³ BLI makes use of the reaction between luciferase enzymes and their substrates, eg, firefly luciferase and luciferin, which produces light.⁴ Clever applications of chemical techniques have allowed bioluminescence to be used to understand

several fundamental mechanistic aspects of cancer biology,¹²⁴ and it is routinely used to monitor the effects of cancer therapy.^{125,126}

Fluorescence, the process of light emission after excitation of a fluorophore with a different wavelength of light, relies on genetically encoded fluorescent proteins or on synthetic or naturally fluorescent molecules, which may be targeted to a cell or protein of interest.⁴ Preclinically, it has found application in the study of protein-protein interactions, cell tracking, and tumor targeting in vivo.^{5–7} The rapidly growing areas of photoacoustic imaging, photodynamic therapy, and photoimmunotherapy all leverage an aspect of fluorescence by detecting sound generated by the thermoelastic expansion of tissues induced by fluorescent light (photoacoustic imaging) or by creating an environment conducive to tumor cell kill (photodynamic therapy and photoimmunotherapy).^{17,18} A significant disadvantage of fluorescence imaging is the intrinsic fluorescence present in normal proteins within tissues, leading to a decrease in signal-to-noise, although this can be addressed through the design of red-shifted fluorescent proteins.⁸ Fluorescent agents that emit in the NIR region (see below) enable sufficient depth of light penetration to allow for real-time surgical guidance, including in clinical trials.⁹

NIR has multiple advantages for intraoperative imaging, including low absorption in blood and other tissues, low scatter, and invisibility to the human eye without the aid of instrumentation.¹⁰ NIR-guided surgery offers opportunities for better discrimination of diseased tissue from normal tissue, decreased margin positivity rates, and minimization of anesthesia times.¹¹ For these reasons, NIR has been extensively explored for guiding cancer surgeries (Fig. 1),¹²⁷ and a specific example is discussed below.

Surface-enhanced Raman scattering (SERS) is another type of optical imaging with high sensitivity and specificity for the delineation of surgical margins. This technique may be an important part of surgical guidance in the future. Jermyn and colleagues studied SERS for intraoperative brain cancer detection.¹²⁸ Those authors reported a sensitivity of 93% and specificity of 91% for the differentiation of normal brain from dense cancer and adjacent brain invaded by cancer cells,¹²⁸ suggesting utility in a class of tumors that is often extensively infiltrative. Li et al used a high-affinity, small-molecule Raman probe targeted against the prostate-specific membrane antigen (PSMA) to selectively identify prostate cancer (PCa) cells,¹²⁹ an important step toward the use of SERS in intraoperative guidance for PCa.

Although optical techniques remain largely in the preclinical domain, advancements in tracer development for

TABLE 1. Selected Modalities for Molecular Imaging in Oncology and Relative Advantages and Disadvantages

MODALITY	USES IN ONCOLOGY	ADVANTAGES/DISADVANTAGES (AVAILABILITY)	SELECTED REFERENCES
Optical			
Fluorescence	Surgical guidance	Nonionizing radiation; photodynamic and photoimmunotherapy/moderate penetration depth (research, translational)	Serkova 2021, ⁴ Rowe & Mapp 2008, ⁵ Rabut & Ellenberg 2004, ⁶ Gao 2010, ⁷ Shaner 2004, ⁸ Hernot 2019, ⁹ Hyun 2016, ¹⁰ Vahrmeijer 2013, ¹¹ Lu 2018, ¹² Ishizawa 2009, ¹³ Xu 2020, ¹⁴ Asanuma 2015, ¹⁵ Hernot 2019 ¹⁶
Photoacoustic	Tissue characterization; surgical guidance	Nonionizing radiation; high optical specificity; real-time/moderate penetration depth (research)	Attia 2019, ¹⁷ Mitsunaga 2011 ¹⁸
Ultrasound	Targeted drug delivery; blood-brain barrier disruption; tumor characterization	Nonionizing radiation; readily available scanners; low cost (translational, clinical)	Xu 2020, ¹⁴ Endo-Takahashi 2020, ¹⁹ Airan 2017, ²⁰ Hult 2020, ²¹ Farhadi 2019 ²²
Magnetic resonance			
Spectroscopy	Brain tumors, prostate cancer	Endogenous contrast; widely available/low sensitivity; limited metabolites (translational, clinical)	Koutcher & Burt 1984, ²³ Pillai 2009, ²⁴ Barker 2009, ²⁵ Negendank 1996, ²⁶ Brindle 2017, ²⁷ Brat 2015, ²⁸ Choi 2012, ²⁹ Li 2015, ³⁰ Chan 2019, ³¹ Edden 2007, ³² Ma 2013, ³³ Claudino 2007, ³⁴ Qi 2019 ³⁵
CEST	Brain tumors, obstructive uropathy	Endogenous contrast; new chemical signatures/technically complex; sensitivity unknown (translational)	Wu 2016, ³⁶ Ward 2000, ³⁷ Xu 2015 ^{38,39}
USPIO	Cell tracking; phagocyte detection; lymph node metastases	Endogenous contrast; new chemical signatures/technically complex; sensitivity unknown (translational)	Zimmer 1995, ⁴⁰ Wu 2019, ⁴¹ Ngen 2021, ⁴² Glover 2020, ⁴³ Mathiasen 2019, ⁴⁴ Schilham 2021, ⁴⁵ Toth 2017, ⁴⁶ Barajas 2019 ⁴⁷
Hyperpolarization	Characterization of tumor metabolism	High signal; potential to investigate a wide array of metabolic pathways; expensive; pathways under study may be perturbed by high concentration of hyperpolarized agents (research)	van Zijl 2021, ⁴⁸ Wang 2019, ⁴⁹ Woitek 2020 ⁵⁰
Radionuclide			
SPECT	Bone scans; brain tumors; sentinel node mapping; radiation dosimetry for theranostics	Widely available/low sensitivity; low resolution; diminishing use in oncology (clinical)	Anger 1952, ⁵¹ Kelly 2020, ⁵² Rowe 2015, ⁵³ Gorin 2016, ⁵⁴ Rowe 2017, ⁵⁵ Wilson 2020 ⁵⁶
PET	Specific molecular targets; metabolism (glucose/glutamine)	High sensitivity and potential for high specificity/complex infrastructure; costly agents (translational, clinical)	Wahl 2008, ⁵⁷ Sanchez-Crespo 2013, ⁵⁸ Verhagen 2021, ⁵⁹ Cherry 2017 ⁶⁰ and 2018, ⁶¹ Zhang 2020, ⁶² Sanli 2017, ⁶³ Wahl 1994, ⁶⁴ Newman 1994, ⁶⁵ Lu 2012, ⁶⁶ Shreve 1999, ⁶⁷ Cheson 2014, ⁶⁸ Hicks 2021, ⁶⁹ Johnson 2016, ⁷⁰ Hope 2019, ⁷¹ Jadvar 2016, ⁷² Fanti 2016, ⁷³ Savir-Baruch 2017, ⁷⁴ Rowe 2019, ⁷⁵ Barratto 2018, ⁷⁶ Eiber 2017, ⁷⁷ Werner 2020, ⁷⁸ Perera 2020, ⁷⁹ Kiess 2015, ⁸⁰ Maurer 2016, ^{81,82} Gorin 2018, ⁸³ Pienta 2021, ⁸⁴ van Leeuwen 2019, ⁸⁵ Cookson 2007, ⁸⁶ Roach 2006, ⁸⁷ Fendler 2019, ⁸⁸ Markowski 2020, ⁸⁹ Morris 2021, ⁹⁰ Joice 2017, ⁹¹ Phillips 2020, ⁹² Sharma 2021, ⁹³ Rohrich 2021, ⁹⁴ Serfling 2021, ⁹⁵ Giesel 2021, ⁹⁶ Lindner 2021, ⁹⁷ Kessler 2021, ⁹⁸ Wu 2021, ⁹⁹ Friedman 2020, ¹⁰⁰ Van Acker 2001, ¹⁰¹ Stumpe 2004, ¹⁰² Familiari 2011, ¹⁰³ Kagna 2012, ¹⁰⁴ Cho 2020, ¹⁰⁵ Mutch 2018, ¹⁰⁶ Ordonez 2021, ¹⁰⁷ Foss 2018, ¹⁰⁸ Petrik 2012 ¹⁰⁹ and 2014, ¹¹⁰ Davies 2017 ¹¹¹
Theranostic	Thyroid cancer, neuroendocrine tumors, prostate cancer	High specificity through image guidance/complex infrastructure; costly agents (translational, clinical)	van der Heil 2003, ¹¹² Strosberg 2017, ¹¹³ Buatti 2021, ¹¹⁴ Burkett 2021, ¹¹⁵ Yadav 2019, ¹¹⁶ Hofman 2018 ¹¹⁷ and 2021, ¹¹⁸ Novartis AG 2021 ¹¹⁹

Abbreviations: CEST, chemical exchange saturation transfer; PET, positron emission tomography; SPECT, single-photon emission computed tomography; USPIO, ultrasmall superparamagnetic iron oxide nanoparticles.

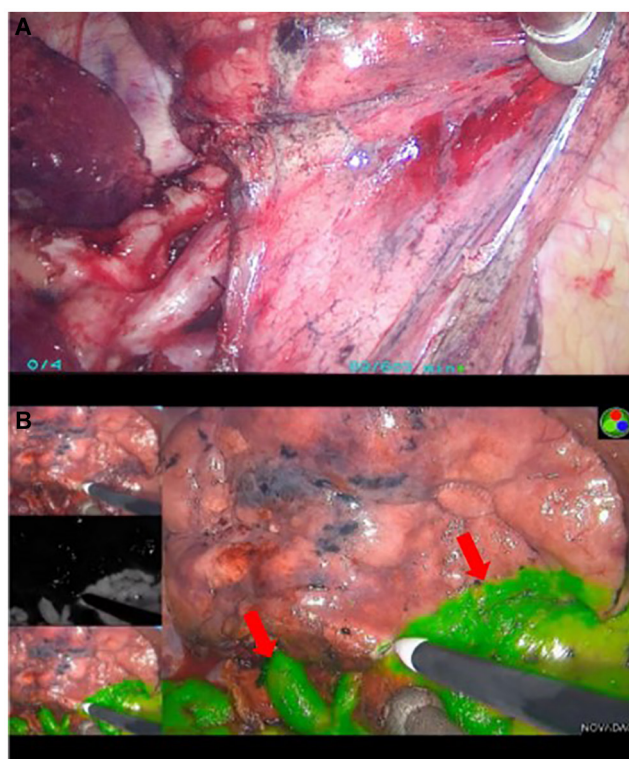


FIGURE 1. The Use of Indocyanine Green for Surgical Guidance During a Lung Segmentectomy. (A) The intersegmental plane was difficult to identify with traditional techniques but (B) was visualized much more clearly with the use of indocyanine green (red arrows in B). Reproduced from: Liu Z, Yang R, Cao H. Near-infrared intraoperative imaging with indocyanine green is beneficial in video-assisted thoracoscopic segmentectomy for patients with chronic lung diseases: a retrospective single-center propensity-score matched analysis. *J Cardiothorac Surg.* 2020;15:303.¹²⁷

other molecular imaging modalities may help to drive the translation of NIR probes into human clinical practice.⁹ Challenges to implementing NIR probes in clinical routine include the need for optimized tracers that have rapid uptake in the tissue of interest but clear quickly from background tissues as well as the intrinsic need for development of highly sensitive instruments and bright fluorescent dyes.⁹ As with many molecular imaging modalities that depend on exogenously administered agents, there are significant barriers to clinical translation, such as expensive biodistribution and toxicology studies that need to be carried out for any new composition of matter.

Magnetic Resonance Imaging

Often classified as an anatomic imaging modality, recent advances with MRI demonstrate the ability of this modality to image molecular processes. All MRI techniques are based on the principle that some atomic nuclei are able to align like small magnets within a magnetic field because of their spin properties.²³ Fundamentally, MRI involves a high magnetic field and the generation of images through the selective application of radiofrequency pulses, which

lead to different patterns of signal in different tissues based on tissue composition, ie, based on the nature and concentration of the nuclei present in those tissues. Traditionally, MRI has been used to create high-resolution anatomic images of soft tissue structures, such as the brain and musculoskeletal system, for which computed tomography (CT) has lacked the contrast resolution to provide useful diagnostic information.

However, MRI uses the same principles as nuclear magnetic resonance spectroscopy (MRS), meaning that it can identify the individual resonances of protons (and other paramagnetic atomic nuclei) and specific compounds if those entities are present in sufficient concentrations. As such, many clinical and investigational MRI techniques fall under the aegis of molecular imaging. For example, MRS can detect compounds that are present at high (millimolar) concentrations and that have a proton signal resolvable from water. As suggested above, MRS uses the same principles of signal acquisition as other MRI techniques. However, the data are analyzed in a different way so that, instead of anatomic images being created, the concentrations of different paramagnetic atoms are displayed as a function of their chemical shift resonances.²⁴ As with other MRI techniques, the massive amount of hydrogen present in biological molecules makes it the paramagnetic atom of choice for MRS, although examining hydrogen atoms in metabolites requires suppression of the signal from hydrogen atoms in surrounding bulk water.

For compounds at lower (micromolar) concentrations, chemical exchange saturation transfer (CEST) can be used, provided the compound of interest has a proton that can be exchanged with surrounding water protons.³⁶ CEST agents were first introduced in 2000 and offer an alternative to traditional MRI contrast materials that increase signal by enhancing water proton relaxivity.³⁷ Although it is able to visualize the presence of substrates at lower concentrations than MRS, CEST still lacks the sensitivity of PET and also can suffer from some of the same specificity issues, including hyperemic effects that may lead to higher signal from exogenously administered agents in inflammation and other conditions. Examples of the uses of MRS and CEST in molecular imaging of cancer are presented below.

An early, preclinical molecular imaging technique that is finding its way into the clinic is the use of ultrasmall iron oxide nanoparticles and other metallic nanoparticles to image phagocytic cells by MRI and, by extension, tumors and metastases with which they become associated.⁴⁰ A further advance of this technology has been the recent development of instrumentation specifically for magnetic particle imaging.⁴¹ Targeted magnetic nanoparticles can serve as a platform to define the depth of penetration of nanoparticles within solid tumors using MRI.⁴² Leveraging the high

signal generated from metallic susceptibility, a key indication for this technology is for cell tracking, including the tracking of transplanted cardiac and other stem cells.^{43,44} In a technique referred to as magnetic resonance (MR) lymphography, ferumoxytol and its analogs have been used to detect lymph nodes involved in PCa, in one clinical instance rivaling PSMA-targeted PET in sensitivity.⁴⁵ Ultrasmall iron oxide nanoparticles have also found substantial application to neuroinflammation⁴⁶ and in tracking pseudoprogression of glioblastoma.⁴⁷

Lastly among the techniques that we will discuss in this section is hyperpolarized MRI, which makes use of a complex process to align the nuclei of ^{13}C -labeled agents to massively increase the signal that is available.^{48,49} Hyperpolarized MRI can be used to investigate a variety of physiologic and pathologic processes, including metabolic pathways in cancer.⁴⁸ A recent example from Woitek et al showed that a reduction in the ^{13}C -labeled lactate-to- ^{13}C -labeled pyruvate ratio was predictive of response to therapy in patients with breast cancer undergoing neoadjuvant chemotherapy.⁵⁰

As with many imaging agents, placing them within a specific environment or changing the isotope if radioactive (see below) can convert them to therapeutics. An imaging agent that, with minimal alteration, can also effect therapy is referred to as a *theranostic*.¹³⁰ For example, placing metallic nanoparticles within an alternating magnetic field creates a heating effect that has proved therapeutic in cancer.¹³¹

Single-Photon Emission Computed Tomography

Although other modalities can provide higher spatial resolution, SPECT remains an important methodology across the gamut of imaging-evaluable pathology. SPECT, like MRI but unlike PET, is clinically ubiquitous. SPECT relies on radiotracers that emit single photons from nuclear decay processes followed by the detection of these photons with a gamma camera. Traditionally, gamma cameras have been composed of a scintillation crystal that converts the emitted photons into visible light,⁵¹ a series of backing photomultiplier tubes that increase the signal from the visible light, and a collimator between the patient and the scintillation crystal that allows the emitted photons to be spatially localized. Gamma cameras can be used for planar imaging; however, in many modern molecular imaging applications, they are spun around the patient to create tomographic images, ie, SPECT.

The limited spatial resolution of SPECT is still adequate for many clinical applications. The fundamental strengths of SPECT derive from the large number of single-photon-emitting radionuclides that are readily available, including technetium-99m ($^{99\text{m}}\text{Tc}$), iodine-123 (^{123}I), and indium-111 (^{111}I). These radionuclides produce emitted photons of different energies, which can be distinguished by the gamma

camera, permitting the simultaneous acquisition of multiple radiotracers. Furthermore, the availability of radionuclides with a long physical half-life ($T_{1/2}$), eg, ^{111}In ($T_{1/2} = 67$ hours), allows for both delayed imaging for diagnostic purposes and the determination of dosimetry for selected therapeutic radiopharmaceuticals.⁵²

Although single-photon-emitting radiotracers lack the high spatial resolution and routine quantifiability of PET radiotracers, the intrinsic advantages of having radionuclides that decay with different energies and the wide array of radiotracers that are available will keep SPECT relevant for routine clinical applications for the foreseeable future.

Positron Emission Tomography

PET is the gold standard for sensitivity in clinical molecular imaging. The basic principle of PET is that proton-rich radionuclides decay by emitting positrons (β^+), which subsequently travel a short distance and annihilate with an electron (β^-) to create two 511-kiloelectron volt photons that arise almost exactly 180 degrees apart.⁵⁷ Rings of detectors can be used to take advantage of coincidence detection to identify the locations of the annihilation events. Common radionuclides used for PET imaging include organic/organic-like isotopes (eg, carbon-11 [^{11}C], nitrogen-13, and fluorine-18 [^{18}F]) and radiometals (eg, gallium-68 [^{68}Ga], copper-64 [^{64}Cu], and zirconium-89 [^{89}Zr]). For many clinical and research applications, ^{18}F provides an ideal combination of medicinal chemistry properties, radionuclide half-life ($T_{1/2} = 110$ minutes), positron yield and energy.⁵⁸

As noted above, radionuclide-based imaging techniques such as PET play important roles in theranostics, namely, in selecting patients for the corresponding therapy. An advantage of radionuclide-based theranostic pairs is that, by merely changing the radionuclide within the chelator, (eg, ^{68}Ga to ^{177}Lu) or changing the isotope of the halogen (eg, ^{123}I to ^{124}I), one may move from an imaging to a therapeutic agent within the same molecular scaffold.

Intrinsic advantages of PET include high-contrast resolution and quantifiable imaging parameters. In modern practice, coregistered CT is used to create attenuation maps that allow highly accurate attenuation correction. With advanced techniques, such as resolution recovery, motion correction, and point-spread function reconstruction, PET is continuing to evolve as a cornerstone of modern clinical molecular imaging. Furthermore, PET is increasingly being combined with MRI (ie, PET/MR), potentially allowing for powerful combinations of the molecular imaging features of each of the individual modalities while also saving radiation dose to patients.⁵⁹ Along those lines, the sensitivity of total-body PET allows for the administration of radiopharmaceutical doses at a fraction of the dose of current clinical studies.^{60–62} That enables expanded use of PET in pediatric populations or for

patients who require frequent studies in whom radiation dosimetry must be carefully taken into account.

Ultrasound

The advantages of ultrasound imaging include real-time dynamic imaging, small physical footprint of the (portable) scanning device, lack of radioactivity, and relatively low cost. Although primarily used clinically for anatomic delineation and for studying flow-based phenomena (Doppler), ultrasound molecular imaging with microbubbles for the targeted delivery of drugs, including genetic material, is proliferating.¹⁹ It is also used for focal disruption of the blood-brain barrier to enable access to the brain for hydrophilic diagnostic and therapeutic agents.²⁰ Photoacoustic imaging may provide highly specific cancer signatures not available from other techniques.²¹ Through modification of what was originally a bacterial gene, Shapiro and coworkers have used ultrasound and an acoustic reporter to image gene expression in mammalian cells.²² The delivery of sound pulses to tissue is now being studied and implemented in analogy to using MR pulse sequences.¹² As more is learned in this area, ultrasound may increase in versatility for biomedical research and medicine through further extension into the molecular realm.

Examples of Molecular Imaging Utilization

The breadth of the impact of modern molecular imaging on medicine and the biomedical sciences is difficult to encapsulate in any brief review. As opposed to a comprehensive listing of current applications of molecular imaging, we present a series of examples that demonstrate how the principles of molecular imaging can affect the care of patients with cancer.

Near-Infrared Imaging for Surgical Guidance

The major limitation of applying optical imaging techniques to human subjects is the limited depth of penetration achievable with the detectors for optical probes. However, this limitation is nearly moot in the context of intraoperative imaging, in which there is exposure of the tissues under study. As such, optical imaging techniques have been studied extensively for the purposes of surgical guidance. NIR probes have been suggested to improve clinical workflow and to have advantages in speed, patient outcomes, and cost relative to traditional, unguided surgical methods.¹¹

The first application of NIR agents to intraoperative guidance made use of indocyanine green, a dye that is approved by the US Food and Drug Administration.²⁰ In an initial study, Ishizawa and coworkers found that hepatobiliary excretion of indocyanine green allowed for the clear delineation of superficial colorectal liver metastases and primary hepatocellular carcinomas.¹³ The tumors were

demarcated by surrounding rims of fluorescence, with little background uptake in the normal liver.

Since that first study, the number of available NIR probes has rapidly expanded to include specific tumor-targeting small molecules, peptides, antibodies, and aptamers.¹⁴ Asanuma and coworkers leveraged the over-expression of β -galactosidase in ovarian cancer so that hydroxymethyl rhodol fluorescence dyes containing β -galactoside could be used to identify peritoneal metastases intraoperatively.¹⁵ Although preclinical, that study is an elegant example of the use of altered patterns of protein expression in cancer to drive specific imaging of subtle sites of cancer dissemination.

NIR surgical guidance is being pursued in a variety of organ systems¹⁶; however, it has been most extensively used to date in the brain. Maximal resection of aggressive gliomas offers advantages in survival, yet it is also important to minimize the impact on surrounding brain to avoid neurological deficits. For these reasons, Butte and coworkers explored the use of the peptide agent Tumor Paint BLZ 100 (Blaze Bioscience, Inc) in mice with implanted human glioma cells.¹³² Those authors reported high signal-to-background in the removed brains from dead mice and believed that the approach was feasible for surgical guidance.¹³² Although a great deal of research remains to be done to have specific, tumor-targeted NIR surgical guidance as part of routine clinical work, the promising preclinical results suggest that this line of inquiry should be broadly pursued. For example, a phase 1 study for the fluorescent identification of positive primary tumor margins and disease-involved lymph nodes in men with PCa is currently accruing patients (ClinicalTrials.gov identifier NCT04574401). In future years, we are likely to see several additional clinical trials for similar indications as new compounds for NIR imaging of novel cancer-relevant targets are developed.

Brain Tumor Imaging with Magnetic Resonance Spectroscopy

The most common metabolites visualized by MRS²⁵ are *N*-acetyl aspartate (NAA), choline (Cho), creatine/phosphocreatine (Cr), and lactate.²⁴ The normal relative concentrations of these compounds in brain are well understood, as are the perturbations in those concentrations in different types of brain tumors. Under normal conditions, NAA, a putative neuronal marker, comprises the highest peak on the spectrum, with lower concentrations of Cr and Cho. Tumors generally demonstrate abnormally decreased NAA/Cr and increased Cho/Cr ratios compared with normal brain, suggesting internal loss of normal neural tissue but also higher rates of cellular turnover; however, intrinsic heterogeneity among tumor grades can make it difficult to derive tumor aggressiveness from MRS data.²⁶

Nonetheless, the clever application of MRS methods to the evaluation of brain tumors has led to the noninvasive determination of important tumor characteristics.²⁷ For example, the discovery of mutant isocitrate dehydrogenase in low-grade gliomas and some glioblastomas that was associated with improved outcomes²⁸ spurred the identification of the oncometabolite 2-hydroxyglutarate by MRS.²⁹ This suggests that noninvasive characterization and prognostication of gliomas may be possible and that tumor heterogeneity that might limit pathological analysis can be overcome with imaging.

Future developments may increase the relevance of MRS. The introduction of ultrahigh-field 7-Tesla scanners into the clinic,³⁰ spectral editing (ie, the targeting of a nucleus or functional group on one species while removing an overlapping resonance due to another species from the spectrum),^{31,32} and the advent of new techniques like MR fingerprinting³³ may overcome some limitations, such as low sensitivity. The complementary field of metabolomics continues to evolve, with potential applications in selection of cancer therapy and response assessment.³⁴ As new patterns of metabolites are identified, MRS is an adaptable modality that can potentially noninvasively identify those patterns. Furthermore, as with other modalities discussed in this review, MRS is likely to benefit from the burgeoning role of artificial intelligence (AI) in radiology. There are already hints that MRS that is leveled-up with machine learning (a type of AI) may overcome the limitations of individual features for noninvasively predicting glioma tumor grade.³⁵ Qi and coworkers found that a machine learning-driven model generated an area under the curve of 0.820 for a validation set, which was better than the individual performance of traditional metabolic features.³⁵

However, these emerging technologies and applications must mature before MRS techniques are likely to be part of routine clinical MR acquisitions in oncology. Nonetheless, noninvasive means of molecular characterization of brain tumors carry considerable promise, and preclinical and translational efforts will continue.

Brian Tumor Imaging With Chemical Exchange Saturation Transfer MRI

In a key preclinical study, Xu and coworkers inoculated mice with a human glioma cell line and then used glucose as a dynamic contrast medium to visualize the tumors.³⁸ By using a frequency offset that could detect the water-exchangeable hydroxyl protons in glucose (glucoCEST), areas of the blood-brain barrier became apparent on the images. Termed dynamic glucose-enhanced (DGE) MRI, this method has significant appeal in that the infused contrast agent is also an endogenous metabolite, and there is no administration of potentially toxic gadolinium metal chelates.

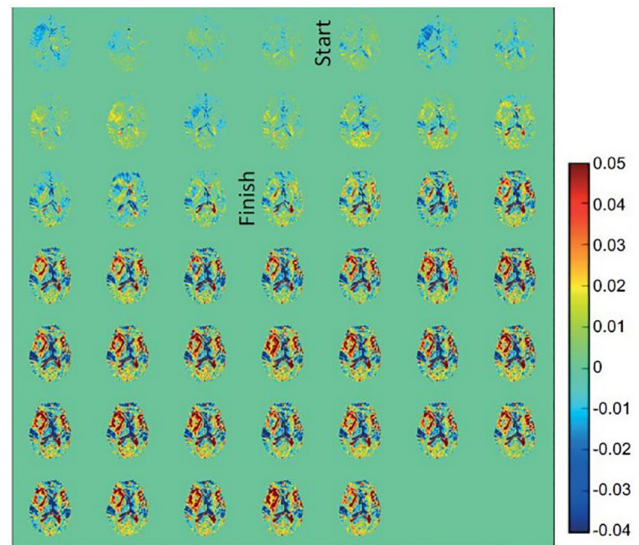


FIGURE 2. A 25-Year-Old Man With Anaplastic Astrocytoma. These are axial dynamic glucose-enhanced difference images at 5.3-second temporal resolution. Note the differential and heterogeneous enhancement centered in the region of the right insula and extending into the right frontal and temporal lobes. Although the contrast between the abnormal right side and the normal left side is less than might be encountered with some other molecular imaging modalities, it is nonetheless impressive that tumor visualization can take place through the exogenous application of glucose. Reprinted from: Xu X, Yadav NN, Knutsson L, et al. Dynamic glucose-enhanced (DGE) MRI: translation to human scanning and first results in glioma patients. *Tomography*. 2015;1: 105–114.³⁹

Given promising preclinical results, DGE MRI has been explored in human subjects.³² In an initial study with 4 healthy volunteers and 3 patients who had glioma, DGE glucoCEST demonstrated spatially variable enhancement within the tumors (Fig. 2).³⁹ Interestingly, the enhancement both varied with time and was not strictly concordant with regions of gadolinium-based enhancement.³⁹ These findings suggest that there may be additional information available from DGE MRI that may include the relative permeability of the blood-brain barrier³⁹ and the degree of inherent glycolytic metabolism within different parts of the tumor. Given the apparent spatially and temporally complex processes governing uptake of glucose and gadolinium-based contrast agents, it is highly likely that insights from AI will be needed to derive detailed prognostic information from those patterns.

Renal Mass Characterization

Multiple modalities, including MRI, SPECT, and PET, have all played a role in bringing molecular imaging to the forefront of indeterminate renal mass characterization. Until recently, renal masses were typically imaged with multiphase, contrast-enhanced anatomic imaging protocols using CT or MRI, and limited information on the nature of the masses could be gleaned from their enhancement patterns.^{133,134} Although renal mass biopsy is a safe and effective means of risk stratification,¹³⁵ the negative predictive value for ruling

out cancer is low, and there is a relatively high nondiagnostic rate.¹³⁶ Because of these limitations, most patients undergo empiric partial or radical nephrectomy on the assumption that most indeterminate renal masses will be renal cancer. This leads to thousands of unnecessary surgeries in the United States each year.¹³⁷

Molecular imaging has provided a means by which indeterminate renal masses may be better evaluated before surgical resection. Ideally, molecular imaging methods would allow for the differentiation of benign/indolent tumors (such as oncocytomas) from aggressive renal cell carcinoma (RCC) subtypes (such as clear cell RCC [ccRCC]). For example, multiparametric MRI that includes functional sequences examining the diffusivity of water and the dynamic contrast enhancement of tissues can be used to create a ccRCC likelihood score (ccLS).¹³⁸ The ccLS tracks with the positive predictive value for a lesion representing a ccRCC, from 5% for ccLS1 through 93% for ccLS5.¹³⁸

SPECT and PET have also been used for renal mass characterization. Multiple different targets have been leveraged. An important example is the development of radiolabeled girentuximab, a monoclonal antibody against carbonic anhydrase IX that is highly expressed on ccRCC cells.¹²⁴ I-girentuximab PET/CT was used in patients with indeterminate renal masses in the REDECT trial (ClinicalTrials.gov identifier NCT00606632) and was found to have better sensitivity and specificity than CT for the identification of ccRCC.¹³⁹

Recently, a commonly used radiotracer for cardiac and parathyroid imaging, ^{99m}Tc-sestamibi, a lipophilic cation that accumulates in accordance with the charge potential of mitochondrial membranes, has been leveraged for imaging indeterminate renal masses.^{53,54} Sestamibi localizes to oncocytomas and other low-grade oncocytic neoplasms on the basis of high mitochondrial content in these lesions, while at the same time it is actively expelled from aggressive RCCs by multidrug resistance pumps.⁵⁵ This inexpensive approach has been adopted across several studies, with meta-analytic sensitivity of 92% and specificity of 88%,⁵⁶ for the identification of renal oncocytomas.

The difficulty of characterizing renal masses is likely to mean that multiple techniques will need to be used for complete nonsurgical characterization. The use of molecular imaging with confirmatory biopsy¹⁴⁰ and emerging genomic approaches,¹⁴¹ if properly used in a risk-stratification approach, can decrease the number of unnecessary renal tumor resections. A proposed workup algorithm, based on¹⁴² and incorporating molecular imaging with ^{99m}Tc-sestamibi SPECT, is shown in Figure 3.^{141,142} This approach is cost-effective, treats the fewest number of benign/indolent tumors, and leaves the least number of aggressive tumors untreated,¹⁴⁰ suggesting high value to patients with indeterminate renal masses.

Cancer Imaging With 2-Deoxy-2-[¹⁸F]Fluoro-D-Glucose

So transformative to oncology imaging has been the widespread clinical adoption of 2-deoxy-2-[¹⁸F]fluoro-D-glucose (¹⁸F-FDG) that entire textbooks have been written on the subject.⁵⁷ ¹⁸F-FDG localizes in most types of malignancy because of its glucose-analog structure, which leads to uptake via GLUT1 transporters in cells undergoing glycolytic metabolism.⁵⁷ The strength of ¹⁸F-FDG is its lack of specificity—the universality of its mechanism of uptake makes it broadly useful for staging, restaging, and therapeutic monitoring of numerous malignancies, from head and neck squamous cell carcinoma⁶³ to bronchogenic carcinomas,⁶⁴ lymphomas,⁶⁵ and myeloma.⁶⁶

However, the weakness of ¹⁸F-FDG PET is also its lack of specificity. ¹⁸F-FDG uptake is seen in numerous nononcologic conditions,⁶⁷ including posttreatment changes (eg, postoperative or postradiation inflammatory change), infections, and granulomatous and nongranulomatous systemic inflammatory processes. Nonetheless, despite these pitfalls, the ability of ¹⁸F-FDG PET to identify the presence of small volumes of malignant disease in morphologically normal structures, as well as to assess the metabolic activity of morphologically abnormal structures, has revolutionized the modern practice of oncology. Perhaps nowhere is this more apparent than in the contemporary approach to the imaging of most Hodgkin and non-Hodgkin lymphomas, in which metabolic assessments on serial ¹⁸F-FDG PET images are a key determinant of response assessment and patient therapy selection (Fig. 4). The Lugano classification is an extensively validated 5-point scale that categorizes posttherapy ¹⁸F-FDG PET scans on the basis of any residual or new metabolic activity and the implications for such activity on the presence of viable lymphoma.⁶⁸

Although new radiotracers are being increasingly used for specific types of cancer, and other agents in early clinical development may challenge the role of ¹⁸F-FDG as the predominant, generalized cancer imaging radiopharmaceutical,⁶⁹ several advances are likely to keep ¹⁸F-FDG highly relevant in years to come. For example, ¹⁸F-FDG PET can be used to adapt therapy in patients with Hodgkin lymphoma. Johnson et al studied a cohort of 1214 patients with newly diagnosed Hodgkin lymphoma.⁷⁰ The authors randomly assigned patients who had negative ¹⁸F-FDG PET scans after 2 cycles of doxorubicin, bleomycin, vinblastine, and dacarbazine (ABVD) to receive either 4 more cycles of ABVD or 4 cycles of AVD (ie, with bleomycin omitted).⁷⁰ They concluded that the use of ABVD led to lower pulmonary toxicities without a significant decrease in efficacy.⁷⁰ One can imagine using ¹⁸F-FDG PET as a means to adapt therapy in a wide variety of different cancers. Furthermore, ¹⁸F-FDG is an inexpensive and readily available radiotracer whose widespread applicability and time-tested clinical

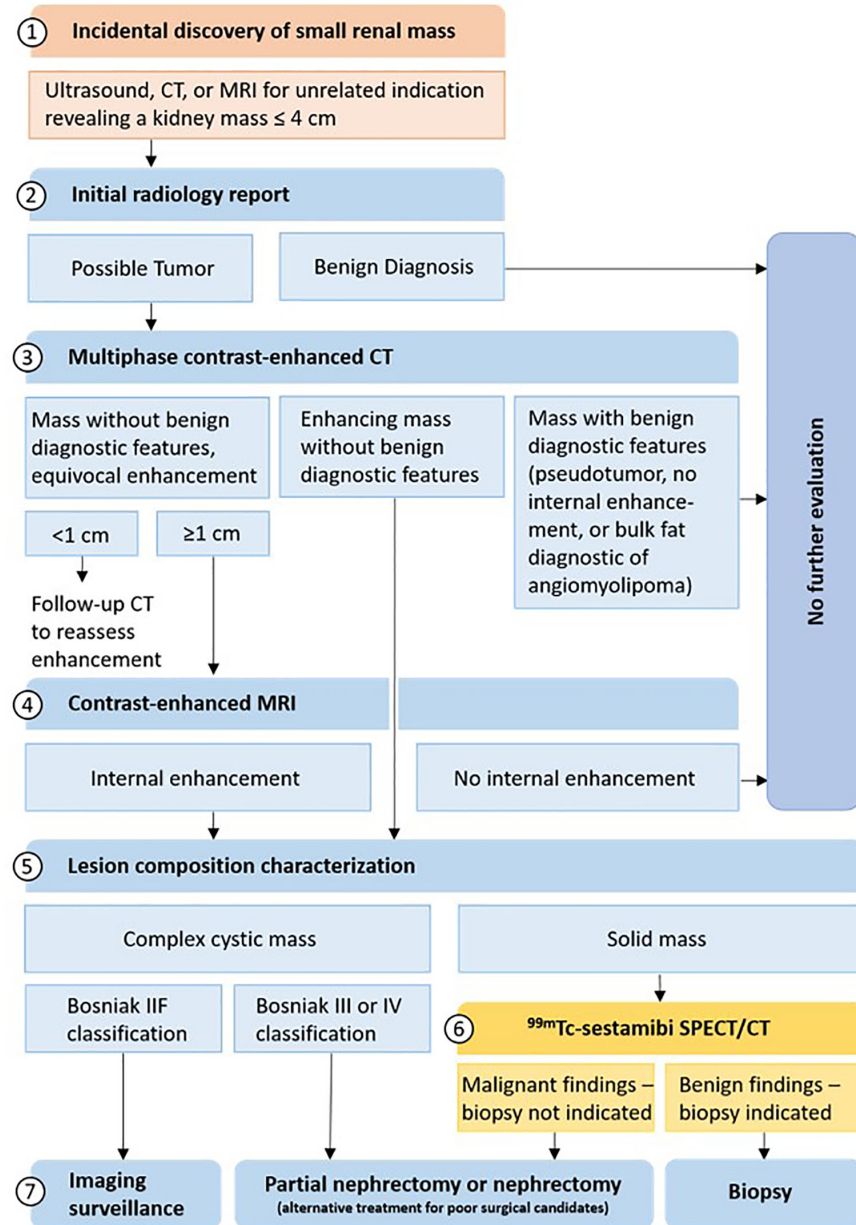


FIGURE 3. Proposed Workup Algorithm for Indeterminate Renal Masses Incorporating Molecular Imaging with Technetium-99m (^{99m}Tc)-Sestamibi Single-Photon Emission Computed Tomography (SPECT). This algorithm is derived from a previously published figure (Kang SK, Bjurlin MA, Huang WC. Management of small kidney tumors in 2019. *JAMA*. 2019;321:1622–1623¹⁴²). CT indicates computed tomography; MRI, magnetic resonance imaging; SPECT, single-photon emission computed tomography.

protocols will make it difficult to displace as a general cancer imaging agent. Other than those cancers that have limited hypermetabolism, the *one-size-fits-all* approach of using ^{18}F -FDG, in which the same patient preparation, dose, uptake time, and scanner protocol can be applied for all cancers, is likely to hold sway over oncology molecular imaging for the near future.

Somatostatin Receptor-Based Imaging and Therapy

Tumors comprised of cells of neuroendocrine origin will often express large amounts of somatostatin receptors

(SSTRs) on their surfaces, particularly SSTR subtype 2. This provides a means of imaging and therapy through the application of high-affinity SSTR ligands. For many years, the SPECT radiotracer ^{111}In -pentetreotide (OctreoScan) was used in this context, generally for imaging,¹⁴³ although, with high doses, therapy was possible.¹¹²

However, the current molecular imaging and therapy paradigm for neuroendocrine tumors is based on the PET radiotracer ^{68}Ga -DOTATATE (or one of several closely related agents) and its therapeutic analog, ^{177}Lu -DOTATATE. Relative to OctreoScan, ^{68}Ga -DOTATATE

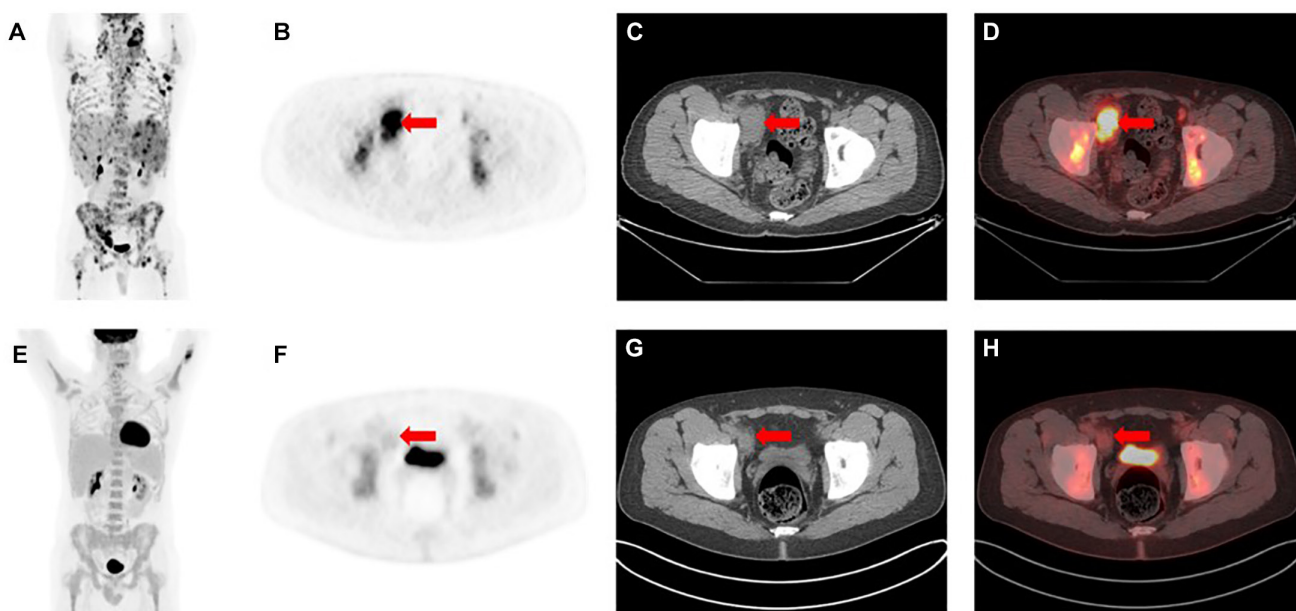


FIGURE 4. A 24-Year-Old Man With Epstein-Barr Virus–Associated Lymphoma Before and After Systemic Therapy. (A) This maximum intensity projection (MIP) 2-deoxy-2-[^{18}F]fluoro-D-glucose (^{18}F -FDG) positron emission tomography (PET) image before the initiation of therapy demonstrates numerous sites of abnormal radiotracer uptake throughout lymph nodes and skeletal structures, consistent with widespread lymphomatous involvement. Axial (B) ^{18}F -FDG, (C) computed tomography (CT), and (D) fused ^{18}F -FDG PET/CT images through the pelvis demonstrate a particularly prominent right external iliac lymph node with intense uptake, consistent with a site of disease (red arrows). (E) In this MIP ^{18}F -FDG PET image after completion of therapy, all of the abnormal uptake has resolved (note that the apparent focus of uptake in the left arm is a result of motion artifact). Axial (F) ^{18}F -FDG, (G) CT, and (H) fused ^{18}F -FDG PET/CT images through the pelvis show that the left external iliac node has decreased in size, although it remains enlarged (red arrows). Despite the residual anatomic abnormality, the uptake has been reduced to blood pool levels, consistent with a complete metabolic response. This example demonstrates the ability of ^{18}F -FDG PET to characterize residual anatomic lesions after therapy.

leverages the advantages of PET (eg, improved spatial and contrast resolution) to produce images with higher sensitivity and higher uptake in sites of disease.⁷¹ Figure 5 is a representative example.

^{177}Lu -DOTATATE was approved by the US Food and Drug Administration for the treatment of patients with well differentiated midgut neuroendocrine tumors after the results of the NETTER-1 phase 3 clinical trial (ClinicalTrials.gov identifier NCT01578239).¹¹³ In NETTER-1, 229 patients were randomized to receive either ^{177}Lu -DOTATATE and best supportive care or long-acting nonradioactive octreotide.¹¹³ Treatment with ^{177}Lu -DOTATATE and best supportive care yielded significant improvements in progression-free survival and the response rate, as well as preliminary evidence of improved overall survival, versus nonradiative octreotide.¹¹³ In the United States, the results of the NETTER-1 trial and subsequent approval of ^{177}Lu -DOTATATE fundamentally changed the approach to the treatment of patients with midgut neuroendocrine tumors, and those patients are now routinely being treated with ^{177}Lu -DOTATATE.

^{177}Lu -DOTATATE has rapidly become the targeted radiopharmaceutical therapy archetype whose routine use is forcing nuclear medicine groups in the United States to adopt a more patient-centered focus, or they risk irrelevance in the current theranostics revolution.¹¹⁴ With PSMA-based imaging and therapies on the horizon of regulatory approval

(see PSMA-Targeted Imaging and Therapy, below), the model in which nuclear medicine physicians play a central role in the multidisciplinary care of patients with metastatic cancer¹¹⁵ is a priority for the field.

PSMA-Targeted Imaging and Therapy

Unlike most malignancies, PCa often does not undergo glycolytic metabolism and is often poorly imaged by ^{18}F -FDG, although there may be times when ^{18}F -FDG is appropriate.⁷² However, the insensitivity of ^{18}F -FDG in many scenarios has led to the development of several molecular imaging agents to image PCa, including those based on imaging metabolic pathways (eg, ^{11}C -choline⁷³ and ^{18}F -fluciclovine⁷⁴) as well as those targeted to specific cell-surface targets (eg, PSMA⁷⁵ and gastrin-releasing peptide receptor⁷⁶).

Multiple radiotracers have received regulatory approval for imaging PCa, although the degree to which these agents have been used in clinical practice has varied widely. However, one of the important contributions of modern molecular imaging has been the development and burgeoning clinical adoption of small-molecule radiotracers that target PSMA for imaging and therapy.⁷⁷ For diagnostic purposes, these agents have generally been labeled with ^{18}F ⁷⁸ or ^{68}Ga ,⁷⁹ allowing high-contrast-resolution PET imaging of sites of PCa.

The clinical applications of PSMA-targeted imaging agents are widely varied, and a complete accounting

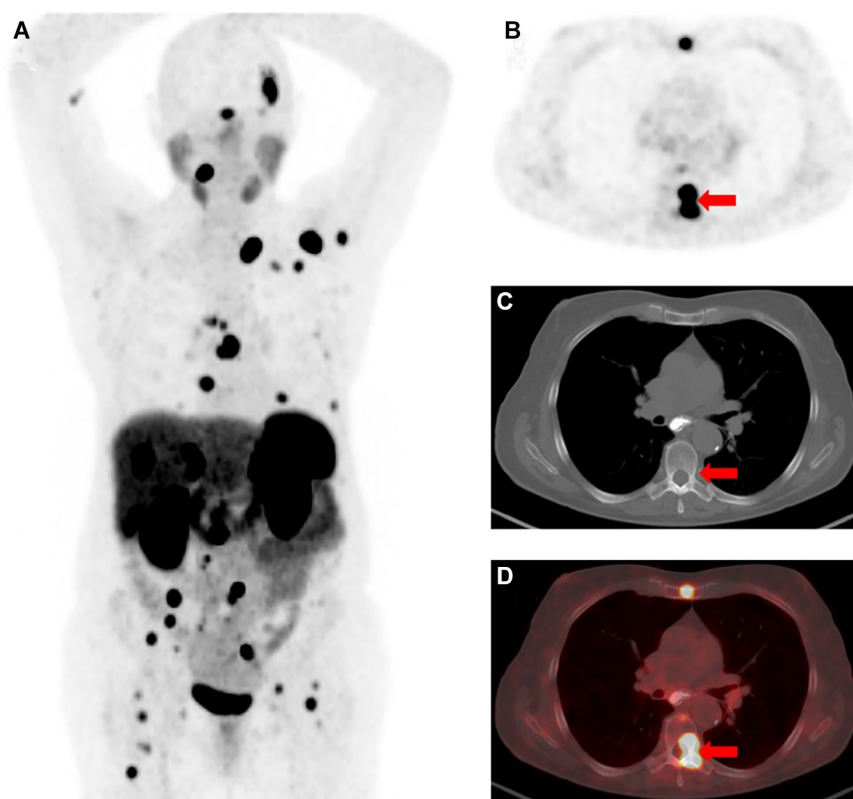


FIGURE 5. A 74-Year-Old Woman With Metastatic Small Bowel Neuroendocrine Tumor. (A) This maximum intensity projection, gallium-68 (^{68}Ga)-DOTATATE, positron emission tomography (PET) image shows numerous sites of abnormal uptake in lymph nodes and bones. Axial (B) ^{68}Ga -DOTATATE PET, (C) computed tomography (CT), and (D) fused ^{68}Ga -DOTATATE PET/CT images demonstrate that many of the bone lesions are easily visible on the PET image but are occult on the corresponding CT anatomic images (red arrows). This case demonstrates the high sensitivity that is achievable with optimized PET radiotracers and that normal-appearing anatomic structures can harbor disease that is well visualized with molecular imaging.

is beyond the scope of the current article. Several detailed reviews have been written, such as those by Rowe et al,⁷⁵ Kiess et al,⁸⁰ and Maurer et al.⁸¹ However, there are applications that merit specific mention here. The first of those indications is primary staging in patients at risk of locoregional nodal or distant metastatic disease. In general, single-center studies^{82,83} have reported higher sensitivity for otherwise occult sites of disease compared with multicenter studies.⁸⁴ In the OSPREY study (ClinicalTrials.gov identifier NCT02981368), for example, 252 patients with high-risk PCa underwent imaging with PSMA-targeted ^{18}F -DCFPyL (a highly selective, small-molecule PET radiotracer) and then proceeded to radical prostatectomy with pelvic lymph node dissection.⁸⁴ The median sensitivity from 3 central readers compared with histopathology was 40.3%, although the very high median specificity (97.9%) led to high median positive and negative predictive values (86.7% and 83.2%, respectively).⁸⁴ Regardless of sensitivity, the presence of visible disease on the scan has prognostic significance because these patients have worse outcomes than those who have *false-negative* scans.⁸⁵

Perhaps the most common indication for a PSMA PET study is in men who had previously attempted curative therapy for their PCa but then had recurrence of a detectable/

rising serum prostate-specific antigen (PSA) level. This state is known as biochemical recurrence (BCR) and is defined by the American Urological Association as a serum PSA level of 0.2 ng/mL in a patient who underwent prior surgery and had achieved an undetectable PSA level⁸⁶ or as a PSA rise of 2.0 ng/mL over nadir in a patient who received prior radiation therapy.⁸⁷ Serum PSA is such a remarkably sensitive means of detecting recurrence that subtle rises suggest the presence of residual or recurrent disease. At low PSA levels, however, the volume of disease may be very small, limiting the value of anatomic imaging in localizing the site of PCa responsible for the rise in PSA. The high-contrast resolution of PSMA PET allows high-sensitivity detection of small foci of disease at low PSA levels.^{88,89} Prospective, multicenter studies have now borne out the use of PSMA PET for localizing recurrent PCa.^{88,90} For example, the detection efficiency was 475 of 635 (75%) at the patient level in a 2-center study using the ^{68}Ga -PSMA-11 radiotracer.⁸⁸ Figure 6 is an example of a patient who had a small-volume recurrence imaged with PSMA-targeted ^{18}F -DCFPyL.

Both in men with BCR and in those with limited volume metastatic disease, ie, oligometastatic PCa,⁹¹ the role of PSMA PET imaging is the localization of disease to potentially allow for nonsystemic curative therapies. In the

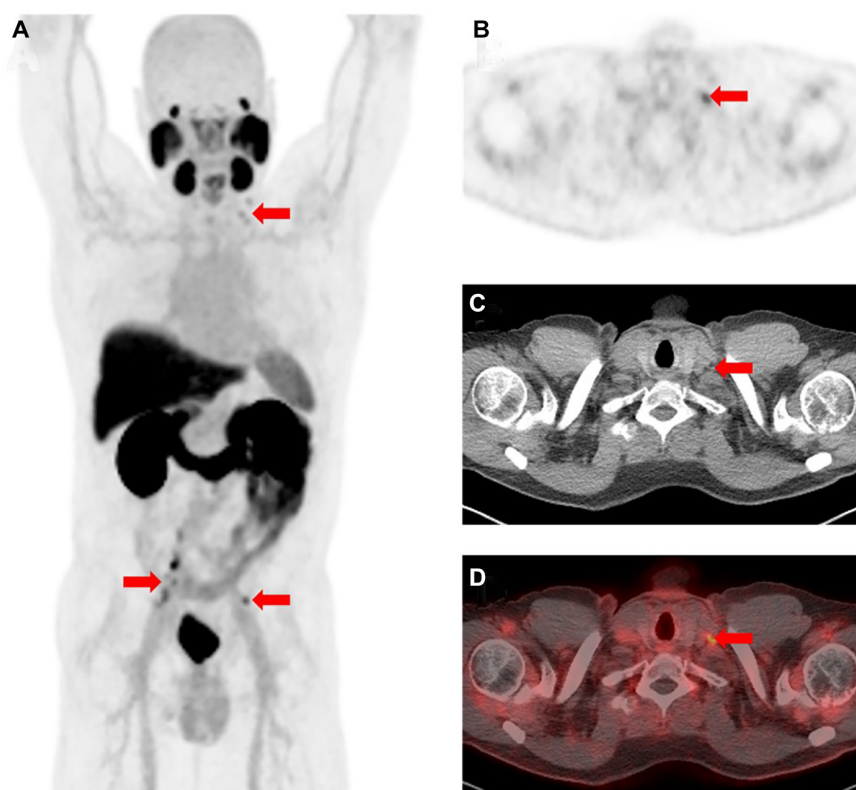


FIGURE 6. A 64-Year-Old Man Who Was 11 Years Postprostatectomy for Gleason 4 + 5 = 9, Grade Group 5 Prostate Cancer and Presented for Prostate-Specific Membrane Antigen (PSMA) Positron Emission Tomography (PET) With a Prostate-Specific Antigen Level of 2.2 ng/mL. (A) This maximum intensity projection, fluorine-18 DCFPyL (^{18}F -DCFPyL) (a highly selective, small-molecule PET radiotracer), PSMA-targeted PET image demonstrates subtle uptake at multiple sites of small, morphologically normal lymph nodes (red arrows). Axial (B) ^{18}F -DCFPyL PET, (C) computed tomography (CT), and (D) fused ^{18}F -DCFPyL PET/CT images demonstrate focal uptake in a 2-mm left supraclavicular (Virchow) node (red arrows), consistent with low-volume systemic nodal disease. Conventional imaging with bone scan and CT had not indicated a site of disease.

ORIOLE trial (ClinicalTrials.gov identifier NCT02680587), men with small-volume metastatic disease were randomized to observation versus stereotactic ablative body radiation (SABR) to visible sites of disease on conventional imaging.⁹² In a post-hoc analysis, it was found that men who had received SABR to all PSMA-avid sites of disease had improved progression-free and distant-metastasis-free survival relative to those whose SABR plan had not included all PSMA-positive disease.⁹²

In men with widespread metastatic PCa, the presence of high uptake of PSMA-targeted PET radiotracers in their sites of disease would suggest that PSMA-based therapeutic molecules may be effective. Although many early studies showing the promise of this approach were retrospective,¹¹⁶ there have been multiple, recent, key prospective trials establishing the effectiveness of ^{177}Lu -labeled agents in treating PSMA-positive PCa. These key studies include the LuPSMA trial (ClinicalTrials.gov identifier NCT04663997), which prospectively demonstrated the efficacy of treatment with ^{177}Lu -PSMA-617¹¹⁷; the TheraP trial (ClinicalTrials.gov identifier NCT03392428), which demonstrated that patients who received ^{177}Lu -PSMA-617 had better oncologic outcomes and less toxicities compared

with those who received cabazitaxel¹¹⁸; and participants in the VISION trial (ClinicalTrials.gov NCT03511664). The VISION trial has recently been shown to meet its primary end points of increased overall survival and time to radiographic progression over best standard-of-care therapy in patients with PSMA-expressing, metastatic, castration-resistant disease.¹¹⁹

With a recent limited US Food and Drug Administration approval of ^{68}Ga -PSMA-11 at 2 sites in the United States¹⁴⁴ and nationwide approval of ^{18}F -DCFPyL,¹⁴⁵ we are on the precipice of PSMA PET imaging revolutionizing the care of men with PCa in the United States. Soon after approval of the imaging agents will likely come the approval of PSMA-based therapeutics, beginning with ^{177}Lu -PSMA-617. Coming years will see a flood of studies to define the utility of PSMA PET imaging, uncover imaging biomarkers for patient prognosis, and understand the role of PSMA therapy in the broader therapeutic landscape for PCa. Already, PSMA-based imaging is a new standard of care for the staging or restaging of men with either primary disease or BCR who are at risk of metastases. The majority of men who are imaged with PSMA PET will undergo a change in treatment plan, indicating profound clinical impact.

Fibroblast-Activation Protein-Based Imaging

Fibroblast-activation protein (FAP) is a rapidly emerging target for imaging and therapy of various cancers. FAP is an excellent target for molecular imaging applications because of its limited normal tissue distribution combined with high expression on cancer-associated fibroblasts.⁹³ To date, the literature on imaging with FAP ligands is primarily retrospective and single-center; however, there are promising potential applications already emerging. For example, pancreatic ductal adenocarcinoma can be difficult to image with conventional anatomic imaging or ¹⁸F-FDG PET. Rohrich et al have reported that a ⁶⁸Ga-labeled inhibitor of FAP (⁶⁸Ga-FAPI) changed the staging in 10 of 19 patients (53%) from a mixed cohort with recurrent/progressive and newly diagnosed disease.⁹⁴

In addition to improved sensitivity for some cancer types that may have low or heterogeneous uptake of ¹⁸F-FDG, the low background uptake of ⁶⁸Ga-FAPI in normal tissues may also imbue ⁶⁸Ga-FAPI PET with added specificity in some circumstances. As one example, Serfling and coworkers recently imaged 8 patients who were suspected of having cancer of unknown primary source within the Waldeyer ring.⁹⁵ Often, ¹⁸F-FDG PET is limited in its ability to discriminate small primary tumors in the Waldeyer ring from background uptake in lymphoid tissue. Although the primary lesions were more visually conspicuous with ⁶⁸Ga-FAPI PET, metastatic lymph node detection was found to be inferior to ¹⁸F-FDG PET.⁹⁵

Given the intrinsic advantages of ¹⁸F-labeled compounds, investigators have begun labeling FAPI-based radiotracers with radiofluorine, including the use of ¹⁸F-aluminium fluoride in place of ⁶⁸Ga in the chelator moieties of existing compounds.⁹⁶ A subset of such radiotracers has been found to have favorable pharmacokinetics with high tumor-to-background ratios.⁹⁷

With burgeoning interest in imaging the tumor micro-environment and the early success of FAP ligands in many clinical studies to date, one can expect this class of radiotracers to be extensively explored for a variety of indications in coming years. Successful clinical acceptance of FAP-targeted PET imaging will depend on prospective studies that prove clinical utility in specific disease states as well as the development of expertise in the imaging community, with imaging specialists being trained to recognize known pitfalls of interpretation.⁹⁸ Figure 7 is an example of FAPI PET imaging in a woman with metastatic lung cancer and demonstrates the ability of FAPI-targeted uptake to be higher than ¹⁸F-FDG uptake in some hypermetabolic cancers and also suggests that FAP-based therapy may be effective in some patients with metastatic disease.⁹⁹

Infection Imaging

Although infection imaging may seem to be a tangent from the primary focus of this review, it is exceptionally

important for patients with underlying cancers. Many patients with cancer are immunocompromised because of the use of chemotherapy, immunotherapy, stem cell transplants, and other treatment modalities. As such, they are susceptible to infections with many organisms, both typical community acquired pathogens and atypical viruses, bacteria, and fungi. Patients with cancer are also frequently imaged, meaning that even subclinical infections may come to light. For these reasons, below, we briefly discuss the emerging role of molecular imaging in characterizing infections.

Anatomic imaging modalities often provide nonspecific information regarding the presence of infection versus sterile inflammation.¹⁴⁶ All types of inflammation, whether infected or sterile, can lead to infiltrative inflammatory changes, edema, and abnormal contrast enhancement.¹⁴⁶ Furthermore, other than a small number of imaging patterns seen with specific pathogens,¹⁴⁷ anatomic imaging does not provide information on the specific causative agent in an infection.

As noted above, ¹⁸F-FDG can have uptake in infectious and inflammatory processes, although the lack of specificity of ¹⁸F-FDG¹⁴⁶ limits its utility for identifying and characterizing infections. To date, no definitive indication for imaging with ¹⁸F-FDG in suspected infection exists, although evaluation of patients with cardiac devices may be a reasonable situation in which to pursue ¹⁸F-FDG.¹⁰⁰ Other scenarios in which complex anatomy can make conventional imaging evaluation difficult—such as musculoskeletal periprosthetic infections^{101,102} and the diabetic foot/Charcot arthropathy^{103,104}—have yielded mixed results with ¹⁸F-FDG PET.

The radically different metabolic pathways possessed by most pathogens relative to the human host suggest that bacteria-specific molecular imaging may be possible. Several radiotracers that leverage unique pathogen metabolism have been investigated,^{105,106} with varying degrees of success. This should remain an active area of investigation, given the importance of identifying pathogens that may otherwise be difficult to obtain through invasive sampling and the impact that proper antibiotic therapy can have on averting morbidity and mortality. For example, Ordonez et al demonstrated that *Enterobacteriales* infections can be specifically imaged with 2-deoxy-2-[¹⁸F]fluoro-D-sorbitol, a radiolabeled sugar derivative that is not used by mammalian cells.¹⁰⁷ Figure 8 is an example of the uptake of 2-deoxy-2-[¹⁸F]fluoro-D-sorbitol in a patient with known *Enterobacteriales* infections and demonstrates the potential utility of this agent to follow response to therapy.¹⁰⁷ The utilization of radiotracers for bacteria-specific metabolic pathways may dovetail with further investigations into agents that have higher specificity for host inflammatory cells than ¹⁸F-FDG.¹⁰⁸

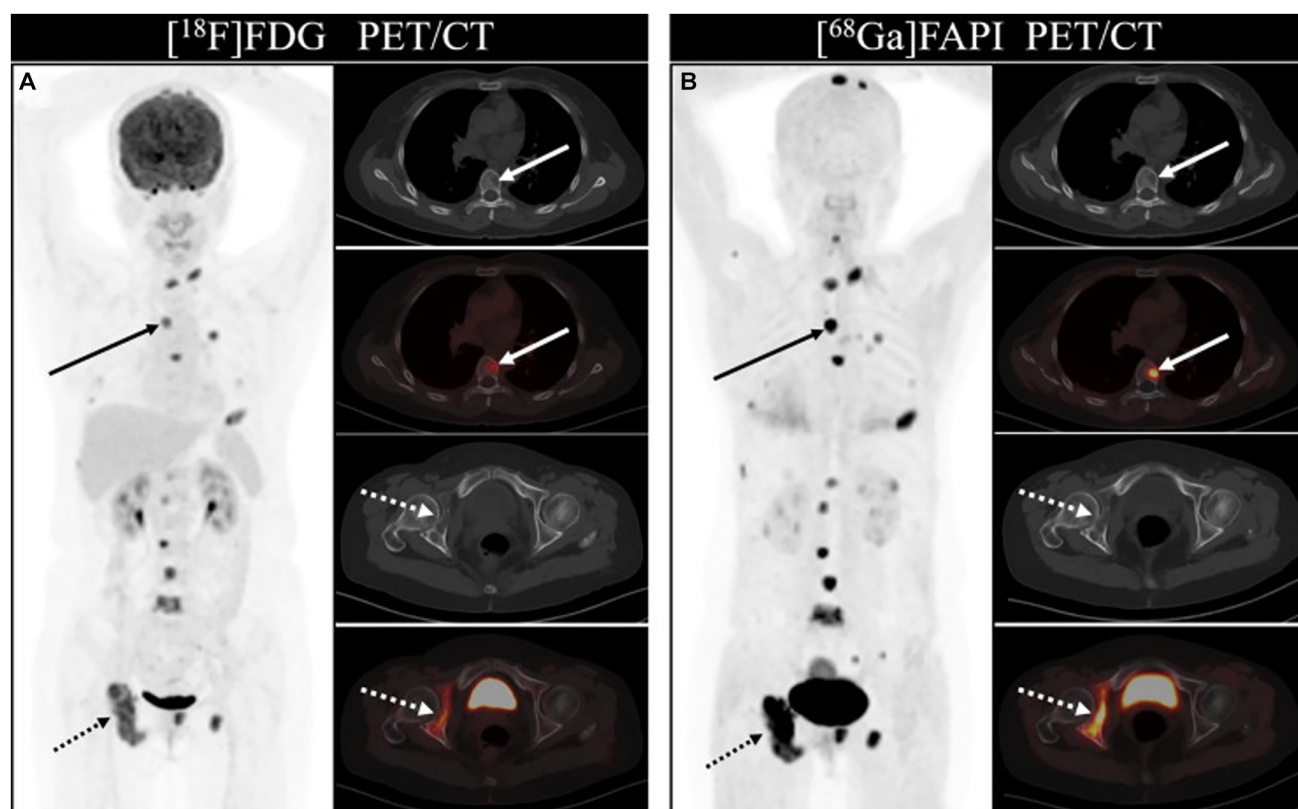


FIGURE 7. As Theranostics Becomes an Ever-Increasing Aspect of Cancer Therapy, Our Ability to Image Relevant Targets Becomes More Important. (A,B) These images are from a 46-year-old woman with newly diagnosed metastatic lung cancer. In B, not only does (right) a gallium-68-labeled inhibitor of fibroblast-activation protein (FAP) positron emission tomography demonstrate higher uptake, but it also suggests that FAP-directed therapy may be effective for some patients with metastatic cancers. Reprinted from: Wu J, Wang Y, Liao T, et al. Comparison of the relative diagnostic performance of [^{68}Ga]Ga-DOTA-FAPI-04 and [^{18}F]FDG PET/CT for the detection of bone metastasis in patients with different cancers. *Front Oncol.* 2021;11:737827.⁹⁹

Limitations to bacteria-specific molecular imaging include a difficult patient population (potentially very sick patients with many underlying comorbidities), clinical workflows that emphasize early administration of antibiotics in patients suspected of having infection, and host factors that may limit blood flow and radiotracer delivery to an area of concern.¹⁰⁵ Whether these limitations can be overcome so that bacteria-specific molecular imaging can provide actionable information to clinicians remains to be seen.

Nonbacterial pathogens can also be evaluated with molecular imaging. For example, fungal uptake of siderophores has been leveraged as a means for imaging clinically relevant fungal organisms, as was demonstrated by Petrik et al in a preclinical model of *Aspergillus fumigatus* infection.¹⁰⁹ Carefully selected siderophores are orthogonal to human physiology and may offer significant specificity advantages relative to metabolic radiotracers.¹¹⁰ Specific imaging of *Aspergillus* species can also be achieved through the use of a humanized monoclonal antibody to the GalF fungal-specific antigen.¹¹¹ The relatively common nature of fungal infections in immunocompromised patients and the inherent difficulties with obtaining and culturing specimens emphasize

the need for noninvasive means of diagnosis. Molecular imaging may provide that, although clinical studies will need to be successfully carried out.

Challenges, Potential, and Future Directions

With a wide array of modalities and numerous probes and techniques available, molecular imaging is not a technologically constrained field. The pace of preclinical discovery tracks far ahead of the rate at which new discoveries are clinically translated. Some of this is related to intrinsic advantages of preclinical work, with the availability of facile techniques such as BLI as well as the quickness with which meaningful oncologic outcomes (eg, overall survival) can be determined. However, there is also an overriding regulatory environment that curtails the effective translation of new molecular imaging agents in the United States.

Generally, in the United States, new radiotracers for SPECT and PET require an Investigational New Drug (IND) application with the US Food and Drug Administration. Such radiotracers are subject to the same regulatory restrictions as therapeutic agents that are administered under an IND. This is despite implementation

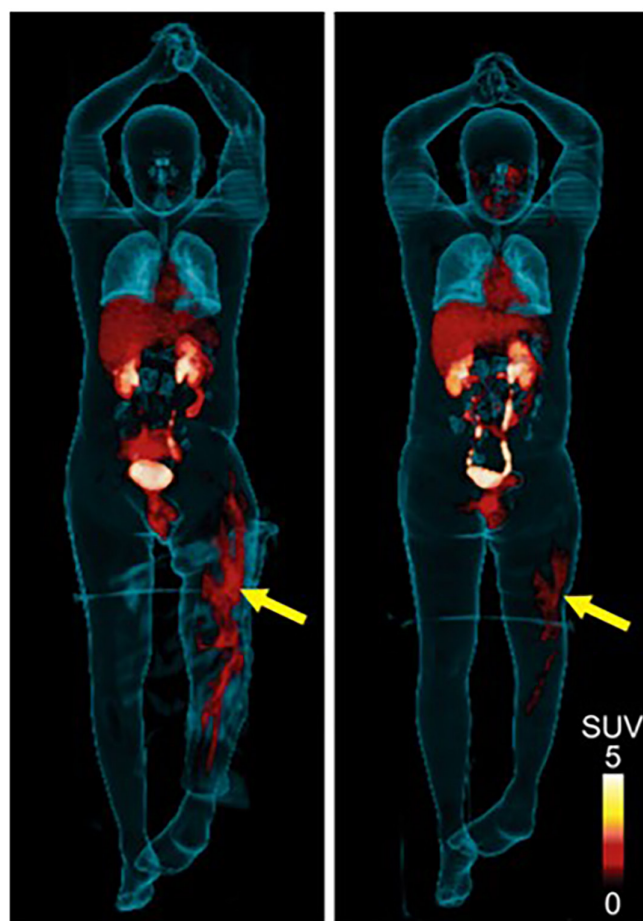


FIGURE 8. Whole-Body 2-Deoxy-2-F Fluorine-18 [^{18}F]Fluoro-D-Sorbitol (^{18}F -FDS) Images of a 33-Year-Old Man Who Had Left Leg Osteomyelitis. (Left) This imaging study was obtained at baseline. (Right) This image was obtained after attempted therapy. The yellow arrows show that uptake at the site of infection decreased but did not resolve; clinically, the patient had persistent infection after therapy. These images suggest that ^{18}F -FDS positron emission tomography may be a means of following and determining the adequacy of antimicrobial therapy. SUV indicates standard uptake value. Reprinted with permission from: Ordonez AA, Wintaco LM, Mota F, et al. Imaging *Enterobacteriales* infections in patients using pathogen-specific positron emission tomography. *Sci Transl Med.* 2021;13:eabe9805.¹⁰⁷ Copyright © 2021 American Association for the Advancement of Science. doi:10.1126/scitranslmed.abe9805

of the tracer principle and the subpharmacologic mass doses that are typically administered to patients. The clinical translation of new diagnostic imaging agents and the execution of phase 1 and 2 studies that would examine human feasibility could be streamlined with a modified regulatory process. Mitigating the onerous requirements for development and submission of an IND would improve the efficiency with which new SPECT and PET radiotracers could be delivered to patients and would also put the United States on a more even footing with other nations as a leader in new radiopharmaceutical innovation. PSMA-targeted PET imaging of PCa for some clinical scenarios had already been incorporated into practice guidelines in Europe¹⁴⁸ before any such agents were approved in the United States. Although prospective clinical

trials in Europe with novel agents or indications are governed by Clinical Trial Applications, in some countries in Europe, compassionate use doctrines can be used to provide access to new molecular imaging agents and support retrospective research studies.

Regardless of the regulatory environment in the United States, novel molecular imaging approaches will continue to be adopted throughout the world as a means of improving the diagnosis and therapy of cancer. Theranostics, specifically, are primed for a rapid expansion in coming years.¹⁴⁹ As noted above, it will be incumbent upon nuclear medicine physicians to lead in ensuring that patients have access to new radiopharmaceutical therapies and that those therapies can be safely and effectively administered in nuclear medicine departments.

The increasing use of theranostic agents for managing cancer will dovetail with the use of AI for several relevant applications. In this context, we will generally be referring to *weak AI*, ie, AI algorithms based on neural networks that can learn from existing data and make relevant and accurate determinations when exposed to new data.¹⁵⁰ With AI for automated whole-body image interpretation, disease segmentation, and burden determination,¹⁵¹ it will be possible to apply the principle of theranostics in powerful ways to improve patient care. Among the many foreseeable applications of the information derived from AI are patient selection for an appropriate theranostic agent, prognostication based on imaging and clinical parameters, and selection of an appropriate dose that balances efficacy with tolerable side effects.

AI can also be used to uncover imaging biomarkers associated with the response of tumors to different treatments. Mu and coworkers used a type of AI known as deep learning to analyze ^{18}F -FDG PET/CT scans to identify features that were associated with epidermal growth factor receptor (EGFR) status.¹⁵² Higher EGFR deep-learning scores (EGFR-DLS) were positively associated with longer progression-free survival intervals for patients who received tyrosine kinase inhibitors targeted to EGFR mutants, whereas EGFR-DLS was negatively associated with multiple outcome measures, including longer progression-free survival intervals, in patients who received immunotherapy.¹⁵²

The utilization of AI-derived imaging biomarkers to guide clinical decision making regarding appropriate therapy will almost certainly expand exponentially in coming years. Characteristics of primary tumors may be leveraged for the prediction of occult metastatic disease, allowing for the appropriate selection of local or systemic therapy. Predictive algorithms will also be developed that will allow for the selection of new therapies for patients undergoing progression or dedifferentiation—before those changes in tumor biology can be appreciated by the human eye.

In short, the confluence of AI and molecular imaging promises to radically alter our approach to imaging in the diagnosis of disease. These 2 fields will influence each other in a synergistic manner. Current molecular imaging agents will continue to be used, generating large data sets that can be leveraged for the development of AI algorithms. The important biological information derived from molecular imaging will be particularly high-yield in driving the ability of AI to generate meaningful clinical outcomes prediction. In turn, AI will drive the development of new molecular imaging radiotracers, will help abstract important information from molecular imaging studies too subtle for human visual detection, and will

provide powerful prognostic information for referring clinicians and patients.

Conclusions

Since the term was coined in the late 1990s, *molecular imaging* has rapidly evolved as a field of tremendous potential to improve the diagnosis and management of patients. Molecular imaging provides information beyond that available from anatomic imaging modalities, allowing for more fundamental insights into pathophysiologic processes. With the concurrent rise of AI and the development of new imaging agents to interrogate novel biological pathways, molecular imaging may soon be among the most important elements in clinical management. ■

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